Rubric: Evolutionary Search for a Domain of Your Choice

Bryn Mawr CS 367: Software Analysis

Weights

Checkpoint 1 (November 15)	15%
Checkpoint 2 (November 29)	15%
Final report (December 18)	70%

Checkpoints

The checkpoints are mainly for feedback. Each is graded as **complete** (full credit), **partial** (half credit), or **missing** (no credit).

Checkpoint 1 is complete if:

- The problem is a real search / optimization problem: fitness is computable and the search space is too large to enumerate
- Every Part 2 design question has a specific answer for your domain
- You explain how you will handle invalid individuals, if your domain has them

Checkpoint 2 is complete if:

- Your GA runs end to end on your problem
- You report results for your default configuration over at least 10 seeds
- You report at least one experiment from Part 4.1, with a fitness-over-generations plot

Final Report

Each criterion is scored from 1 to 5. A criterion's contribution to the report grade is its weight $\times$ (score / 5).

Criterion	**5**	**4**	**3**	**2**	**1**
Problem formulation (10%)	Clearly explains the domain, what is optimized, and why the search space is too large to enumerate.	Domain and fitness are clear; the case for search is brief but convincing.	Domain and fitness are clear, but the case for search is thin.	Fitness is vague, or the search space is small.	Unclear what is optimized, or the problem can be solved by brute force.
GA design (10%)	Every component is specific to the domain and justified. Invalid individuals are handled with a clear rationale.	All components are specific to the domain; one or two choices lack justification.	Components are described, but several choices are not justified.	Some components are generic or don't fit the representation.	Components are missing or don't fit the representation.
Experimental methodology (20%)	Default configuration is justified. One parameter changes at a time. At least 10 seeds per configuration. Budgets are comparable.	Sound setup with one minor gap.	Experiments are run, with gaps such as too few seeds or uneven budgets.	Several parameters change at once, or one required experiment is missing.	Single runs, or several required experiments missing.
Results (20%)	Every experiment has a labeled fitness-over-generations plot and a table with the mean and standard deviation. Easy to compare.	All results present; a few plots or tables are missing labels or standard deviations.	Results present, but several are hard to read or compare.	Some results missing or not tied to a configuration.	Most results missing or unreadable.
Random search comparison (10%)	Random search creates individuals the same way the GA creates its initial population, and gets the same number of fitness evaluations. Clear comparison, well explained.	Fair setup; explanation is brief.	Comparison present, but random search gets a different number of fitness evaluations, or the explanation is thin.	Random search is set up in a way that favors one side, or the comparison is hard to interpret.	Missing.
Discussion (20%)	Explains *why* results came out as they did, with evidence. Answers the discussion questions directly. Honest about negative results.	Explains most results with evidence; one question answered thinly.	Describes what happened with some explanation of why.	Mostly restates results with little interpretation.	No interpretation, or claims the data doesn't support.
Writing (10%)	Organized, concise, and easy to follow. Within the page limit.	Clear, with minor organization issues.	Mostly clear, with some confusing sections.	Often hard to follow, or outside the page limit.	Very hard to follow.